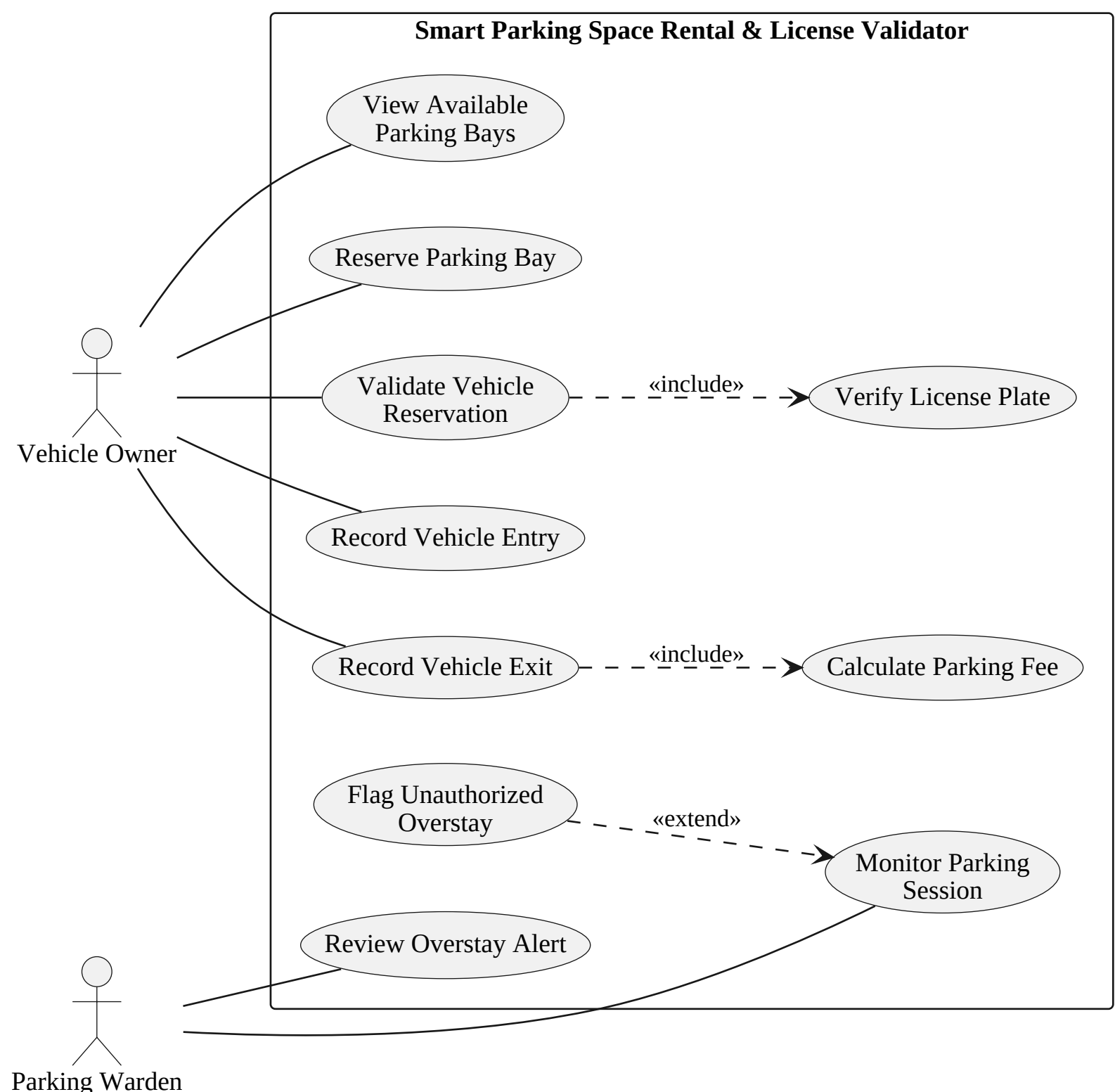
Requirements Table

Smart Parking Space Rental & License Validator

ID	Type	Description	Priority	Acceptance Criteria	Rationale
FR-001	Functional	The system shall allow a Vehicle Owner to view currently available parking bays and reserve an available bay for a selected parking period.	High	**Pass:** The selected available bay is successfully reserved and is no longer shown as available for the reserved period. **Fail:** The same bay can be simultaneously reserved by multiple users.	Real-time reservation prevents parking conflicts and improves parking-space utilization.
FR-002	Functional	The system shall capture the vehicle license plate at the parking entrance and validate it against an active reservation before permitting entry.	High	**Pass:** A vehicle with a valid matching reservation is successfully verified. **Fail:** A vehicle without a valid matching reservation is approved.	License-plate validation ensures that only authorized vehicles use reserved parking spaces.
FR-003	Functional	The system shall record the vehicle entry time when a valid vehicle enters the parking facility and record the exit time when the vehicle leaves.	High	**Pass:** Correct entry and exit timestamps are stored for the parking session. **Fail:** Entry or exit timestamps are missing or incorrectly recorded.	Accurate timestamps are required for session tracking, billing, and overstay detection.
FR-004	Functional	The system shall calculate the total parking fee upon vehicle exit based on the elapsed parking duration and the applicable progressive hourly pricing tier.	High	**Pass:** The calculated amount correctly corresponds to the recorded parking duration and applicable pricing tier. **Fail:** An incorrect duration or pricing tier is used.	Accurate progressive billing ensures that Vehicle Owners are charged correctly.
FR-005	Functional	The system shall monitor active parking sessions and flag vehicles that remain parked beyond their authorized reservation period for review by the Parking Warden.	High	**Pass:** A vehicle exceeding its authorized parking period is flagged for the Parking Warden. **Fail:** An unauthorized overstay is not detected or flagged.	Automatic overstay detection allows Parking Wardens to handle unauthorized parking efficiently.
NFR-001	Performance & Security	The gate-camera license-plate lookup and reservation-validation operation shall complete within 300 milliseconds under normal and simulated peak-load conditions.	High	**Pass:** Validation completes within 300 ms while maintaining required security checks. **Fail:** Validation exceeds 300 ms or bypasses security checks.	Fast and secure validation prevents delays and unauthorized entry at the parking gate.
NFR-002	Reliability	The system shall preserve confirmed parking reservations and recover normal reservation-validation functionality within 60 seconds after a temporary service interruption.	Medium	**Pass:** Confirmed reservations remain intact and validation functionality is restored within 60 seconds. **Fail:** Reservations are lost or recovery exceeds 60 seconds.	Reliable recovery minimizes parking-service disruption and prevents loss of reservation information.


Smart Parking Space Rental & License Validator
View Available Parking Bays
Reserve Parking Bay
Validate Vehicle Reservation
«include»
Verify License Plate
Record Vehicle Entry
Record Vehicle Exit
«include»
Calculate Parking Fee
Flag Unauthorized Overstay
«extend»
Monitor Parking Session
Review Overstay Alert
Vehicle Owner
Parking Warden

Use-Case Flow Specification

Smart Parking Space Rental & License Validator

Use Case ID	UC-03
Use Case Name	Validate Vehicle Reservation
Primary Actor	Vehicle Owner
Supporting Actor	Parking Warden
Goal	Validate the arriving vehicle against an active parking reservation before allowing entry.

Preconditions

1. The vehicle has arrived at the parking facility entrance.
2. The gate camera and license-plate validation service are operational.
3. The parking reservation database is accessible.

Postconditions

1. For a valid reservation, the vehicle entry time is recorded.
2. An active parking session is created and the vehicle is permitted to enter.

Main Success Scenario

1. The Vehicle Owner arrives at the parking entrance.
2. The system captures the vehicle license plate using the gate camera.
3. The system reads the detected license-plate number.
4. The system searches for an active reservation associated with the license plate.
5. The system verifies that the reservation is valid for the current date and time.
6. The system records the vehicle entry time.
7. The system activates the corresponding parking session.
8. The system permits the vehicle to enter the parking facility.

Alternate Flow — A1: Invalid or Missing Reservation

1. At Step 4 or Step 5, the system determines that no valid active reservation exists.
2. The system marks the vehicle-entry validation as unsuccessful.
3. The vehicle is flagged for review.
4. The Parking Warden is notified of the failed validation.
5. Entry is denied unless the Parking Warden performs an authorized manual review.